

Harikrishnan Prasannam

Senior Software Engineer - Full Stack

Tamilnadu, India [+919080053714](tel:+919080053714) prasannam.harikrishnan@gmail.com <https://www.hari-krishnan.com>

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Summary

Senior Full Stack Developer with 4+ years of experience in **software development**, specializing in **microservices architecture**, **DevOps practices**, and **cross-platform development**. Proven track record in leading projects, building scalable web and mobile applications, and improving performance and deployment efficiency. Adept in **technical leadership**, **system architecture**, and **cloud computing**, with a strong focus on **code quality**, **user experience design**, and **security best practices**

Skills

Frontend Development

React, Javascript, Typescript, D3.js, Redux, Tailwind CSS

Backend Development

Python, Flask, Express, NodeJS

Middleware Development

GraphQL, REST APIs

Database

MongoDB, ChromaDB(Vector), Neo4j(Graph), PostgreSQL

Cloud & Deployment

Kubernetes (K8s) - Helm, Docker, AWS - EC2, AWS - S3, AWS - Lambda, Azure - VM, Azure Devops, Azure Key Vault

CI/CD & DevOps

Jenkins, Azure Pipeline, Continuous Integration, Dynamic application configurations

Other Tools

Hashicorp Vault, Keycloak, Git (version control), Agile Methodologies, Unit/Regression/E2E Testing

Software Design

Software design patterns, Technical documentation, Scalability solutions, Requirements analysis

Experience

Accenture India Pvt Ltd

Application Development Senior Analyst,

Coimbatore (Remote)

June 2024 - Present

- **Led a Proof of Concept (PoC) to fine-tune a Large Language Model (LLM)** using LoRA (Low-Rank Adaptation), achieving a 30% improvement in accuracy and contextual relevance—demonstrating strong capabilities in **performance optimization** and domain-specific model adaptation.
- **Designed and developed an interactive Knowledge Graph visualization** for advanced data analysis using **D3.js**, **Neo4j**, and AI agents, enabling dynamic path discovery and semantic relationship mapping—applying **software design patterns** to ensure scalability and maintainability.
- **Built an AI-powered documentation generation agent** that extracts insights from source code and produces high-level technical documentation through configurable workflows—enhancing **technical documentation**, developer onboarding, and codebase transparency.
- **Collaborated directly with clients** to analyze requirements and implement tailored add-on features, ensuring alignment with business goals and use cases—demonstrating strong skills in **requirements analysis**, **user experience design**, and **cross-functional communication**.

Accenture India Pvt Ltd

Application Development Analyst

Coimbatore (Remote)

June 2022 - June 2024

- **Implemented Micro Frontend (MFE) architecture using React**, breaking down a monolithic application into independently deployable modules—enabling **parallel development**, improved scalability, and enhanced maintainability across teams through strategic **software design patterns**.
- **Led the complete migration from monolithic to Micro Frontend (MFE) architecture**, deploying modular apps on **AWS** and **Azure** using **Docker**, **Kubernetes (Helm)**, and CI/CD pipelines. Took end-to-end ownership of the **DevOps lifecycle**, including boilerplate setup, **Dockerfile creation**, pipeline scripting, and **Helm chart development**, driving **continuous integration** and deployment efficiency.
- **Designed and architected a distributed worker system** using **Redis**, **WebSockets**, and **Kubernetes Jobs** to handle resource-intensive tasks concurrently—significantly enhancing system responsiveness and **scalability**, while applying **performance optimization** techniques for high-load processing.
- **Developed dynamic dashboards using Recharts**, implementing various interactive chart types and enabling users to **create and configure new charts** within customizable dashboards—enhancing **data visualization**, **user experience design**, and self-service analytics capabilities.
- **Collaborated with platform administration teams** to onboard MFE child applications into a centralized UI framework, ensuring seamless integration, **cross-platform compatibility**, and adherence to **organizational standards** and best practices.

Accenture India Pvt Ltd

Application Development Associate

Chennai (Remote)

May 2021 - May 2022

- **Collaborated with cross-functional teams** to deliver core application features and resolve high-impact bugs—demonstrating adaptability, **agile methodologies**, and effective **debugging** skills in a fast-paced development environment.
- **Designed and implemented a Python-based middleware microservice** for managing **dynamic application configurations**, with secure integration to **Vault providers** (e.g., HashiCorp, Azure Vault), reinforcing **security best practices** and supporting flexible deployment strategies.
- **Developed and enforced Role-Based Access Control (RBAC)** within a **multi-tenant architecture**, leveraging **Keycloak** for identity and access management (IAM)—ensuring secure, scalable user and permission management across clients.
- **Improved API performance by 10x** using advanced **MongoDB aggregation** techniques—resulting in significantly reduced response times and enhanced **scalability** for high-throughput operations.
- **Engineered a dynamic ETL pipeline** using **JMESPath** and **regex-based pattern matching**, enabling real-time data extraction and transformation across diverse and inconsistent data schemas—illustrating strong problem-solving and **software design pattern** application.

Education

M.Kumarasamy College of Engineering

Bachelors of Engineering - Electronics & Communications

GPA - 7.6 | Graduated | May 2021

Projects

All projects can be found at the following link:

<https://www.hari-krishnan.com/#projects>

Certifications

GENAI Practitioner

Accenture

Feb 2024

DevOps Essentials

Accenture

Jan 2022

Awards

Inspiring Innovator Award

Accenture

Dec 2023

Accenture Celebrate Excellence winner in FY23 Q4, Accenture

The Extra Mile Award

Accenture

Mar 2023

Accenture Celebrate Excellence winner in FY23 Q1, Accenture

Languages

English

Tamil